

34.5 Deviance and asymptotic inference

Parts II and III rested on exact distribution theory: $\hat{\boldsymbol{\beta}}$ was normal because $\mathbf{Y}$ was, and sums of squares were chi-squared because they were quadratic forms in a normal vector (Theorem 4.3.2). None of that survives. This section states what replaces it in the limit, under which conditions, and — just as important — where the approximations fail.

34.5.1 Asymptotic normality

The conditions below are stated for the canonical link, where the argument can be given in full. Throughout $\boldsymbol{\beta}^0$ is the true parameter, $\theta_i^0 = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}^0$, and

$$\mathbf{A}_n = \mathcal{I}_n(\boldsymbol{\beta}^0) = \frac{1}{\phi} \mathbf{X}^\top \text{diag}\{w_i b''(\theta_i^0)\} \mathbf{X}$$

is the expected information from Eq. 34.3.2. Because the canonical-link Hessian does not involve $\mathbf{y}$ (Theorem 34.3.1(d)), $\mathbf{A}_n$ is a fixed matrix, not a random one. Define the scalars

$$q_{ni} = \frac{w_i b''(\theta_i^0)}{\phi} \mathbf{x}_{(i)}^\top \mathbf{A}_n^{-1} \mathbf{x}_{(i)}, \quad i = 1, \dots, n. \quad (34.5.1)$$

These are the exact analogue of leverages: nonnegative, summing to $\text{tr}(\mathbf{A}_n^{-1} \mathbf{A}_n) = p$, so they average p/n as in Proposition 6.8.1(b). The conditions are

- (G1) the responses are independent, the model is that of Definition 34.2.1 with the canonical link, $\text{rank}(\mathbf{X}) = p$ with p fixed, and ϕ and the w_i are known;
- (G2) $\boldsymbol{\beta}^0$ is interior to $\mathcal{B}$, the natural parameters θ_i^0 lie in a fixed compact subset K of the interior of Θ , and $0 < w_{\min} \leq w_i \leq w_{\max} < \infty$;
- (G3) $\max_{i \leq n} q_{ni} \rightarrow 0$: no single observation carries a fixed share of the information;
- (G4) $\lambda_{\min}(\mathbf{A}_n) \rightarrow \infty$: information accumulates in every direction.

Theorem 34.5.1 — Consistency and asymptotic normality of the estimate

Under (G1)–(G4), with probability tending to one the likelihood equations have a solution $\hat{\boldsymbol{\beta}}_n$; it is the unique maximizer of ℓ , it is consistent for $\boldsymbol{\beta}^0$, and

$$\mathbf{A}_n^{1/2}(\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}^0) \xrightarrow{d} N_p(\mathbf{0}, \mathbf{I}). \quad (34.5.2)$$

The same holds with $\mathbf{A}_n$ replaced by $\hat{\mathbf{A}}_n = \mathcal{I}_n(\hat{\boldsymbol{\beta}}_n)$. Consequently, for a fixed $\boldsymbol{\lambda} \neq \mathbf{0}$,

$$\frac{\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_n - \boldsymbol{\lambda}^\top \boldsymbol{\beta}^0}{(\boldsymbol{\lambda}^\top \hat{\mathbf{A}}_n^{-1} \boldsymbol{\lambda})^{1/2}} \xrightarrow{d} N(0, 1),$$

and $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_n \pm z_{1-\alpha/2}(\boldsymbol{\lambda}^\top \hat{\mathbf{A}}_n^{-1} \boldsymbol{\lambda})^{1/2}$ is an interval with asymptotic coverage $1 - \alpha$.

One result is imported, not proved: the standard corollary of **the convexity lemma** of Pollard (1991, section 6), that pointwise convergence of concave random functions is automatically uniform on compact sets. Let Λ_n be random concave functions on $\mathbb{R}^p$, allowed the value $-\infty$ outside a convex set, of the form $\Lambda_n(\mathbf{s}) = \mathbf{s}^\top \mathbf{z}_n - \frac{1}{2} \|\mathbf{s}\|^2 + r_n(\mathbf{s})$ with $\mathbf{z}_n = O_p(1)$ and $\sup_{\|\mathbf{s}\| \leq M} |r_n(\mathbf{s})| \rightarrow 0$ in probability for every M . Then a maximizer $\hat{\mathbf{s}}_n$ exists with probability tending to one and $\hat{\mathbf{s}}_n - \mathbf{z}_n \rightarrow \mathbf{0}$ in probability. Concavity does the work that compactness conditions do in the usual theory, and Theorem 34.3.3 supplies it free.

Proof. Step 1: the score is asymptotically normal. By Eq. 34.3.3, $\mathbf{U}_n = \phi^{-1} \sum_i w_i (Y_i - \mu_i^0) \mathbf{x}_{(i)}$ is a sum of independent mean-zero vectors with $\text{Cov}(\mathbf{U}_n) = \mathbf{A}_n$. Put $\mathbf{z}_n = \mathbf{A}_n^{-1/2} \mathbf{U}_n$. For a fixed unit vector $\mathbf{v}$ write $\mathbf{v}^\top \mathbf{z}_n = \sum_i c_{ni} \xi_i$ with $\xi_i = Y_i - \mu_i^0$ and the weights $c_{ni} = \phi^{-1} w_i \mathbf{v}^\top \mathbf{A}_n^{-1/2} \mathbf{x}_{(i)}$. This is the weighted sum of Lemma 21.2.1, and its case (b) applies. Moments: by (G2) the cumulants of Y_i are continuous functions of θ_i^0 on a compact set (Proposition 34.1.2(c)) and the w_i are bounded away from 0 and ∞ , so $E \xi_i^4$ is bounded — case (b) with $\delta = 2$ — and $\tau_{\min}^2 \leq \text{Var} \xi_i = \phi b''(\theta_i^0)/w_i \leq \tau_{\max}^2$ for constants free of i and n . Weights: by construction $\sum_i c_{ni}^2 \text{Var} \xi_i = \mathbf{v}^\top \mathbf{A}_n^{-1/2} \mathbf{A}_n \mathbf{A}_n^{-1/2} \mathbf{v} = 1$, so the sum is already normalized and $\|\mathbf{c}_n\|^2 \geq \tau_{\max}^{-2}$; while $c_{ni}^2 \leq \phi^{-2} w_{\max}^2 \|\mathbf{A}_n^{-1/2} \mathbf{x}_{(i)}\|^2$ and $\|\mathbf{A}_n^{-1/2} \mathbf{x}_{(i)}\|^2 = \mathbf{x}_{(i)}^\top \mathbf{A}_n^{-1} \mathbf{x}_{(i)} = \phi q_{ni} / \{w_i b''(\theta_i^0)\} \leq C' q_{ni}$ by Eq. 34.5.1 and (G2) again. Hence $\max_i c_{ni}^2 / \|\mathbf{c}_n\|^2 \leq C'' \max_i q_{ni} \rightarrow 0$ by (G3), which is Eq. 21.2.2. So $\mathbf{v}^\top \mathbf{z}_n \rightarrow N(0, 1)$ for every unit $\mathbf{v}$, and the Cramér–Wold device gives $\mathbf{z}_n \rightarrow N_p(\mathbf{0}, \mathbf{I})$.

Step 2: the log-likelihood is asymptotically quadratic. Put $\Lambda_n(\mathbf{s}) = \ell(\boldsymbol{\beta}^0 + \mathbf{A}_n^{-1/2} \mathbf{s}) - \ell(\boldsymbol{\beta}^0)$. Taylor’s theorem with integral remainder, applied to the smooth function ℓ , gives

$$\Lambda_n(\mathbf{s}) = \mathbf{s}^\top \mathbf{z}_n - \int_0^1 (1-t) \mathbf{s}^\top \mathbf{A}_n^{-1/2} \mathcal{J}_n(\boldsymbol{\beta}_t) \mathbf{A}_n^{-1/2} \mathbf{s} dt,$$

where $\boldsymbol{\beta}_t = \boldsymbol{\beta}^0 + t \mathbf{A}_n^{-1/2} \mathbf{s}$ and $\mathcal{J}_n(\boldsymbol{\beta}) = \phi^{-1} \mathbf{X}^\top \text{diag}\{w_i b''(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})\} \mathbf{X}$ is the *nonrandom* observed information. Fix $M > 0$ and let $\|\mathbf{s}\| \leq M$. Then $|\mathbf{x}_{(i)}^\top \mathbf{A}_n^{-1/2} \mathbf{s}| \leq M \|\mathbf{A}_n^{-1/2} \mathbf{x}_{(i)}\| \leq M(C' q_{ni})^{1/2}$, which by (G3) tends to 0 uniformly in i . So for large n every perturbed natural parameter $\theta_i^0 + t \mathbf{x}_{(i)}^\top \mathbf{A}_n^{-1/2} \mathbf{s}$ lies in a fixed compact neighbourhood of K inside Θ , on which b'' is uniformly continuous and bounded away from 0 and ∞ ; hence $b''(\theta_i^0 + \delta_i) = b''(\theta_i^0) \{1 + \epsilon_{ni}\}$ with $\sup_i |\epsilon_{ni}| \rightarrow 0$ uniformly in $t \in [0, 1]$ and $\|\mathbf{s}\| \leq M$, and $\mathbf{A}_n^{-1/2} \mathcal{J}_n(\cdot) \mathbf{A}_n^{-1/2} = \mathbf{I} + \mathbf{R}_n$ with $\|\mathbf{R}_n\| \rightarrow 0$, and

$$\Lambda_n(\mathbf{s}) = \mathbf{s}^\top \mathbf{z}_n - \frac{1}{2} \|\mathbf{s}\|^2 + r_n(\mathbf{s}), \quad \sup_{\|\mathbf{s}\| \leq M} |r_n(\mathbf{s})| \rightarrow 0. \quad (34.5.3)$$

Step 3: the maximizer. By Theorem 34.3.3 each Λ_n is concave. It is defined only on the convex set $\mathbf{A}_n^{1/2}(\mathcal{B} - \boldsymbol{\beta}^0)$, a half-space for the canonical links of the gamma and inverse Gaussian; extend it by $-\infty$ outside, which preserves concavity, and note that Step 2 put every ball $\|\mathbf{s}\| \leq M$ inside that set for large n , so the expansion holds where the lemma needs it. Then Eq. 34.5.3 is exactly the form the lemma requires, with $\mathbf{z}_n = O_p(1)$ by Step 1 and a nonrandom remainder bound, so a maximizer $\hat{\mathbf{s}}_n$ exists with probability tending to one and $\hat{\mathbf{s}}_n - \mathbf{z}_n \rightarrow \mathbf{0}$ in probability. Since $\hat{\mathbf{s}}_n = \mathbf{A}_n^{1/2}(\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}^0)$, Slutsky’s theorem and Step 1 give Eq. 34.5.2, and Theorem 34.3.3 makes the maximizer unique.

Consistency. $\|\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}^0\| \leq \lambda_{\min}(\mathbf{A}_n)^{-1/2} \|\hat{\mathbf{s}}_n\|$, and $\|\hat{\mathbf{s}}_n\| = O_p(1)$, so (G4) gives $\hat{\boldsymbol{\beta}}_n \rightarrow \boldsymbol{\beta}^0$ in probability.

Plug-in. Consistency and the uniform continuity of b'' used in Step 2 give

$$\mathbf{A}_n^{-1/2} \hat{\mathbf{A}}_n \mathbf{A}_n^{-1/2} \rightarrow \mathbf{I}$$

in probability, whence $\hat{\mathbf{A}}_n^{1/2}(\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}^0) \rightarrow N_p(\mathbf{0}, \mathbf{I})$. The last two displays follow by taking the $\boldsymbol{\lambda}$ component. ■

Remark (What the conditions mean, and the non-canonical case). Condition (G3) is the honest content of “large sample”. It is the generalized-linear-model version of the requirement $\max_i h_{ii} \rightarrow 0$ that Chapter 21 needed, and Step 1 used it in exactly that role, to supply Eq. 21.2.2. It fails when it should: a group with a single observation, a regressor that indicates one case, a binary design with a nearly empty cell. The canonical link is used twice, for the concavity and for the nonrandom Hessian, and the next proposition records what survives without it.

Warning. Nothing here is a finite-sample statement: $\hat{\boldsymbol{\beta}}$ is biased, by $O(n^{-1})$ against a standard error of $O(n^{-1/2})$ — asymptotically negligible, not zero. Nor does p fixed cover many parameters: a model with a parameter per group and bounded group sizes violates (G3), and Example (Many means, one variance) showed what goes wrong.

Intervals for a fitted mean. For fixed $\mathbf{x}_0$ the theorem with $\boldsymbol{\lambda} = \mathbf{x}_0$ gives $\hat{\eta}_0 \pm z_{1-\alpha/2}(\mathbf{x}_0^\top \hat{\mathbf{A}}_n^{-1} \mathbf{x}_0)^{1/2}$ for $\eta_0 = \mathbf{x}_0^\top \boldsymbol{\beta}^0$, the counterpart of Theorem 12.1.1 with a normal quantile for a t one; applying the monotone h to both endpoints makes it an interval for $\mu_0 = h(\eta_0)$ of the same asymptotic coverage. The *interval* transforms, not the endpoint formula, so it is not symmetric about $\hat{\mu}_0$ — a feature, since a log link then keeps it positive and a logit link inside $(0, 1)$. The delta-method alternative $\hat{\mu}_0 \pm z_{1-\alpha/2} h'(\hat{\eta}_0)(\mathbf{x}_0^\top \hat{\mathbf{A}}_n^{-1} \mathbf{x}_0)^{1/2}$ agrees to first order and is the worse wherever h curves sharply. Neither is a prediction interval: Theorem 12.3.1 has no clean analogue here.

Proposition 34.5.2 — Non-canonical links

Let the link be any twice continuously differentiable g with $g' \neq 0$ on $\mathcal{M}$, and put $\mathbf{A}_n = \phi^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{X}$ with $\mathbf{W}$ the working weights of Theorem 34.3.1 evaluated at $\boldsymbol{\beta}^0$. Then every conclusion of Theorem 34.5.1 — existence with probability tending to one, consistency, the limit Eq. 34.5.2, the plug-in version and the Wald intervals — continues to hold, as do Proposition 34.5.3 and Theorem 34.5.5(a),(b) below, which use the theorem only through the quadratic expansion Eq. 34.5.3. The conditions are those of Fahrmeir and Kaufmann (1985): the divergence condition $\lambda_{\min}(\mathbf{A}_n) \rightarrow \infty$, and a continuity condition asking that $\mathbf{A}_n(\boldsymbol{\beta}^0)^{-1/2} \mathbf{A}_n(\boldsymbol{\beta}) \mathbf{A}_n(\boldsymbol{\beta}^0)^{-1/2} \rightarrow \mathbf{I}$ uniformly over $\boldsymbol{\beta}$ in neighbourhoods of $\boldsymbol{\beta}^0$ that shrink in the information metric. The two together play the part of (G2)–(G4).

Remark. This proposition is quoted, not proved: without concavity and a nonrandom Hessian the argument above collapses, and the replacement is the content of Fahrmeir and Kaufmann (1985). It is recorded because most models in the chapters that follow use a non-canonical link somewhere — the complementary log–log and probit links of Chapter 35, the gamma/log fit of Example (Food expenditure: three models, two decisions), the negative binomial with a log link in Chapter 37.

34.5.2 Wald, score and likelihood ratio

Consider a reduced model $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ with $\text{rank}(\mathbf{X}_0) = p - q$, and let $\tilde{\boldsymbol{\beta}}$ be the maximum likelihood estimate under it, written as a vector in $\mathbb{R}^p$. Since $\mathbf{X}$ has full column rank, $\{\boldsymbol{\beta} : \mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0)\}$ is a subspace of dimension $p - q$, so it is $\{\boldsymbol{\beta} : \boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{0}\}$ for some $p \times q$ matrix $\boldsymbol{\Lambda}$ of rank q , in the notation of Definition 11.3.1.

Proposition 34.5.3 — The three statistics

Define

$$\text{Wald} = (\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}})^\top (\boldsymbol{\Lambda}^\top \hat{\mathbf{A}}_n^{-1} \boldsymbol{\Lambda})^{-1} \boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}},$$

$$\text{Score} = \mathbf{U}(\tilde{\boldsymbol{\beta}})^\top \mathcal{I}_n(\tilde{\boldsymbol{\beta}})^{-1} \mathbf{U}(\tilde{\boldsymbol{\beta}}),$$

$$\text{LR} = 2\{\ell(\hat{\boldsymbol{\beta}}) - \ell(\tilde{\boldsymbol{\beta}})\}.$$

Under (G1)–(G4) and the reduced model, each of the three converges in distribution to $\chi^2(q)$, and any two of them differ by $o_p(1)$. For a single coefficient ($q = 1$, $\boldsymbol{\Lambda} = \mathbf{e}_j$) the Wald statistic is $\hat{\beta}_j^2 / [\hat{\mathbf{A}}_n^{-1}]_{jj}$, the square of the “estimate over standard error” ratio that every program prints.

Proof. Work in the coordinates $\mathbf{s} = \mathbf{A}_n^{1/2}(\boldsymbol{\beta} - \boldsymbol{\beta}^0)$ of the proof of Theorem 34.5.1, where $\boldsymbol{\beta}^0$ is the true parameter, which lies in the reduced model, so $\boldsymbol{\Lambda}^\top \boldsymbol{\beta}^0 = \mathbf{0}$. The reduced model is the linear subspace $\mathcal{S}_n = \mathbf{A}_n^{1/2} \mathcal{N}(\boldsymbol{\Lambda}^\top) = \mathcal{N}(\mathbf{B}_n)$ of $\mathbb{R}^p$, where $\mathbf{B}_n = \boldsymbol{\Lambda}^\top \mathbf{A}_n^{-1/2}$ has rank q ; let $\mathbf{P}_n$ be the orthogonal projection onto it, so that $\mathbf{I} - \mathbf{P}_n = \mathbf{B}_n^\top (\mathbf{B}_n \mathbf{B}_n^\top)^{-1} \mathbf{B}_n$.

For the likelihood ratio, Eq. 34.5.3 gives $\Lambda_n(\mathbf{s}) = \mathbf{s}^\top \mathbf{z}_n - \frac{1}{2} \|\mathbf{s}\|^2 + r_n(\mathbf{s})$ uniformly on compacts. Maximizing $\mathbf{s}^\top \mathbf{z}_n - \frac{1}{2} \|\mathbf{s}\|^2$ over $\mathbb{R}^p$ gives $\frac{1}{2} \|\mathbf{z}_n\|^2$ at $\mathbf{s} = \mathbf{z}_n$; over $\mathcal{S}_n$ it gives $\frac{1}{2} \|\mathbf{P}_n \mathbf{z}_n\|^2$ at $\mathbf{s} = \mathbf{P}_n \mathbf{z}_n$. Both maximizers are $O_p(1)$, so the remainder is negligible at both, and

$$\text{LR} = 2\left\{ \max_{\mathbb{R}^p} \Lambda_n - \max_{\mathcal{S}_n} \Lambda_n \right\} = \|(\mathbf{I} - \mathbf{P}_n) \mathbf{z}_n\|^2 + o_p(1).$$

If $\mathbf{z} \sim N_p(\mathbf{0}, \mathbf{I})$ then $\|(\mathbf{I} - \mathbf{P}) \mathbf{z}\|^2 \sim \chi^2(q)$ for every projection $\mathbf{P}$ of rank $p - q$ (Theorem 4.3.2). Projections of a fixed rank form a compact set, so along any subsequence $\mathbf{P}_n \rightarrow \mathbf{P}$, and the continuous mapping theorem gives $\|(\mathbf{I} - \mathbf{P}_n) \mathbf{z}_n\|^2 \rightarrow \chi^2(q)$; the limit is the same for every subsequence, so the whole sequence converges.

For the Wald statistic, write $\hat{\mathbf{s}}_n = \mathbf{A}_n^{1/2}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^0)$. Then $\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} = \boldsymbol{\Lambda}^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^0) = \mathbf{B}_n \hat{\mathbf{s}}_n$ and $\boldsymbol{\Lambda}^\top \hat{\mathbf{A}}_n^{-1} \boldsymbol{\Lambda} = \mathbf{B}_n \mathbf{B}_n^\top$, so with $\mathbf{A}_n$ in place of $\hat{\mathbf{A}}_n$ the statistic is exactly $\|(\mathbf{I} - \mathbf{P}_n) \hat{\mathbf{s}}_n\|^2$. The proof of Theorem 34.5.1 gives $\hat{\mathbf{s}}_n = \mathbf{z}_n + o_p(1)$, and $\mathbf{A}_n^{-1/2} \hat{\mathbf{A}}_n \mathbf{A}_n^{-1/2} \rightarrow \mathbf{I}$ in probability makes the substitution of $\hat{\mathbf{A}}_n$ cost $o_p(1)$; hence $\text{Wald} = \|(\mathbf{I} - \mathbf{P}_n) \mathbf{z}_n\|^2 + o_p(1)$. The same argument inside the reduced model gives $\mathbf{A}_n^{1/2}(\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta}^0) = \mathbf{P}_n \mathbf{z}_n + o_p(1)$, which is used next.

For the score statistic, differentiating Eq. 34.5.3 — legitimate because pointwise convergence of differentiable concave functions forces convergence of their gradients (Rockafellar 1970, section 25) — gives $\mathbf{A}_n^{-1/2} \mathbf{U}(\boldsymbol{\beta}^0 + \mathbf{A}_n^{-1/2} \mathbf{s}) = \mathbf{z}_n - \mathbf{s} + o_p(1)$ uniformly on compacts; at $\mathbf{s} = \mathbf{P}_n \mathbf{z}_n + o_p(1)$ this is $(\mathbf{I} - \mathbf{P}_n) \mathbf{z}_n + o_p(1)$. ■

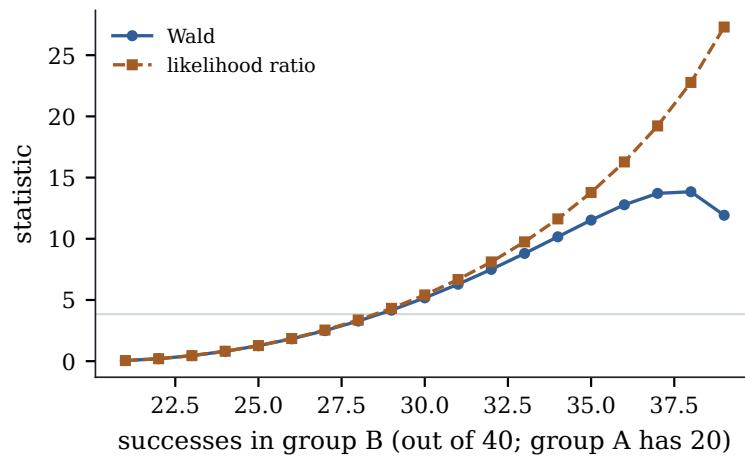


Figure 34.5.1: The Hauck–Donner effect. Two groups of 40 trials, with 20 successes in group A. As the successes in group B increase, the likelihood ratio statistic rises monotonically, while the Wald statistic turns over and falls. The horizontal line is the upper 5% point of $\chi^2(1)$.

In the normal linear model the three are *exact* monotone functions of one another and of the F statistic, with $\text{Wald} > \text{LR} > \text{Score}$ in every sample (Proposition 11.4.2). Outside it that ordering is not guaranteed, and the three agree only to first order.

The Wald statistic is not invariant. Testing $\beta_j = 0$ and testing $e^{\beta_j} = 1$ are the same hypothesis with different Wald statistics, because the delta-method standard error of $e^{\hat{\beta}_j}$ is not the transform of that of $\hat{\beta}_j$; the likelihood ratio statistic, which compares maximized likelihoods, is invariant under every reparameterization.

The Wald statistic can move the wrong way. This failure has a name.

Example. (The Hauck–Donner effect)

Two groups of 40 trials, with 20 successes in group A and s_B in group B, fitted by a logistic model with an intercept and a group indicator. The coefficient is the log odds ratio $\hat{\beta}_1 = \log\{s_B/(40-s_B)\}$, with standard error $\{1/20+1/20+1/s_B+1/(40-s_B)\}^{1/2}$. As s_B rises from 21 to 39 the evidence against equality strengthens monotonically and the likelihood ratio statistic records it, climbing to 27.301. The Wald statistic does not: it rises to 13.842 at $s_B = 38$ and then *falls* to 11.924 at $s_B = 39$, though the fitted log odds ratio has grown to 3.664. The standard error grows faster than the coefficient.

At complete separation, $s_B = 40$, coefficient and standard error are both infinite and the Wald statistic is 0: the test accepts the null precisely where the data support it most strongly (Figure 34.5.1). The effect occurs whenever the log-likelihood is far from quadratic over the range where the estimate lies.

```
import numpy as np

m_hd, s_a = 40, 20                                     # group A: 20 successes out of
40
s_b = np.arange(21, 40)                                # group B: from 21 to 39
successes
odds_ratio = (s_b / (m_hd - s_b)) / (s_a / (m_hd - s_a))
beta1 = np.log(odds_ratio)                             # the fitted log odds ratio
se1 = np.sqrt(1 / s_a + 1 / (m_hd - s_a) + 1 / s_b + 1 / (m_hd - s_b))
```

```

wald = (beta1 / se1) ** 2

def binom_ll(s, m, p):
    return s * np.log(p) + (m - s) * np.log(1 - p)

p_pool = (s_a + s_b) / (2 * m_hd)
lr = 2 * (binom_ll(s_a, m_hd, s_a / m_hd) + binom_ll(s_b, m_hd, s_b / m_hd)
          - binom_ll(s_a, m_hd, p_pool) - binom_ll(s_b, m_hd, p_pool))

for k in (0, 10, 15, 17, 18):
    print(f"successes in B = {s_b[k]:2d}: log odds ratio {beta1[k]:6.3f}"
          f"    Wald {wald[k]:7.3f}    likelihood ratio {lr[k]:7.3f}")

```

[Run this code online](#)

When the log-likelihood *is* close to quadratic, the three agree closely. For the age coefficient of Example (Fitted votes by party identification) the script obtains the single-coefficient Wald = 2.046, Score = 2.054 and LR = 2.066, all far from the 5% point 3.84 of $\chi^2(1)$, so the three lead to the same conclusion.

The score statistic needs only the null fit, so it can test a term from the model without it; but it evaluates the information at $\hat{\beta}$, the wrong place when the alternative is far away. The recommendation is simple: report likelihood ratio tests, and where an interval is wanted invert them, $\{\beta_j : \text{LR}(\beta_j) \leq \chi^2_{1-\alpha}(1)\}$, rather than using the Wald interval.

34.5.3 The deviance

Definition 34.5.1 — Deviance and scaled deviance

Let $\hat{\mu}$ be the fitted means of a model and let $\theta(\cdot)$ be as in Definition 34.1.2. Assuming every $y_i \in \mathcal{M}$, the **saturated model** is the one that sets $\mu_i = y_i$. The **deviance** of the fit is

$$D(\mathbf{y}, \hat{\mu}) = \sum_{i=1}^n d_i, \quad (34.5.4)$$

$$d_i = 2w_i \left[y_i \{ \theta(y_i) - \theta(\hat{\mu}_i) \} - b \{ \theta(y_i) \} + b \{ \theta(\hat{\mu}_i) \} \right],$$

and the **scaled deviance** is $D^* = D/\phi = 2\{\ell_{\text{sat}} - \ell(\hat{\mu})\}$.

The deviance does not involve ϕ ; the scaled deviance does. For the normal family, $\theta = \mu$ and $b(\theta) = \theta^2/2$ give $d_i = w_i(y_i - \hat{\mu}_i)^2$: the deviance is the weighted residual sum of squares, the scaled deviance SSE/σ^2 .

Family	d_i/w_i
normal	$(y - \hat{\mu})^2$
binomial (y a proportion)	$2 \left[y \log \frac{y}{\hat{\mu}} + (1 - y) \log \frac{1 - y}{1 - \hat{\mu}} \right]$
Poisson	$2 \left[y \log \frac{y}{\hat{\mu}} - (y - \hat{\mu}) \right]$

Family	d_i/w_i
gamma	$2\left[-\log \frac{y}{\hat{\mu}} + \frac{y - \hat{\mu}}{\hat{\mu}}\right]$
inverse Gaussian	$\frac{(y - \hat{\mu})^2}{y \hat{\mu}^2}$

with the convention $0 \log 0 = 0$.

Proposition 34.5.4 — The deviance is nonnegative

Every $d_i \geq 0$, with $d_i = 0$ if and only if $\hat{\mu}_i = y_i$. Consequently $D \geq 0$, with equality only for the saturated fit, and D decreases when a model is enlarged.

Proof. Write $\tilde{\theta} = \theta(y_i)$, $\hat{\theta} = \theta(\hat{\mu}_i)$, so $b'(\tilde{\theta}) = y_i$. Then

$$\frac{d_i}{2w_i} = y_i(\tilde{\theta} - \hat{\theta}) - b(\tilde{\theta}) + b(\hat{\theta}) = b(\hat{\theta}) - b(\tilde{\theta}) - b'(\tilde{\theta})(\hat{\theta} - \tilde{\theta}) \geq 0$$

by convexity of b (Lemma 34.1.1): the right-hand side is the Bregman divergence of b , the amount by which b lies above its tangent at $\tilde{\theta}$. Strict convexity makes it strict unless $\hat{\theta} = \tilde{\theta}$, that is $\hat{\mu}_i = y_i$. Enlarging the model can only raise the maximized likelihood, and $D = 2\phi\{\ell_{\text{sat}} - \ell(\hat{\mu})\}$. ■

The other standard discrepancy is the **Pearson statistic**,

$$X^2 = \sum_{i=1}^n \frac{w_i(y_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)}, \quad (34.5.5)$$

which is $\sum_i (y_i - \hat{\mu}_i)^2 / \widehat{\text{Var}}(Y_i)$ up to the factor ϕ . It equals the deviance for the normal family and approximates it elsewhere, since a second-order expansion of d_i about $\hat{\mu}_i = y_i$ gives $w_i(y_i - \hat{\mu}_i)^2 / V(\hat{\mu}_i)$.

34.5.4 Analysis of deviance

Theorem 34.5.5 — Distribution theory for the deviance

Assume (G1)–(G4).

- (a) (*Nested models.*) Let $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ with $\text{rank}(\mathbf{X}) - \text{rank}(\mathbf{X}_0) = q$. Under the reduced model,

$$\frac{D_0 - D}{\phi} \xrightarrow{d} \chi^2(q),$$

where D_0 and D are the deviances of the two fits. When ϕ is unknown it is replaced by an estimate, and the ratio $\{(D_0 - D)/q\}/\hat{\phi}$ is referred to $F(q, n - p)$, by analogy with Theorem 11.1.1.

- (b) (*Grouped data.*) Suppose the data consist of G groups, G and p fixed, with y_i an average of $w_i = m_i$ observations and $m_i \rightarrow \infty$ for every i . Then under the fitted model $D/\phi \rightarrow \chi^2(G-p)$ and $X^2/\phi \rightarrow \chi^2(G-p)$, so both are goodness-of-fit statistics.
- (c) (*Ungrouped data.*) If the number of groups grows with the sample size, (b) does not apply and the limit is generally not $\chi^2(n-p)$. For binary responses with the canonical link the failure is total: D is a function of $\hat{\beta}$ alone, so it carries no information about fit beyond what $\hat{\beta}$ already contains.

Proof.

- (a) By Definition 34.5.1 the saturated log-likelihood cancels in the difference, so $(D_0 - D)/\phi = 2\{\ell(\hat{\beta}) - \ell(\tilde{\beta})\}$, which is the likelihood ratio statistic of Proposition 34.5.3.
- (b) Regard the data as the $\sum_i m_i$ individual observations rather than the G averages, and take the saturated model to be the one with a separate parameter per group, so that its model matrix is the group indicators and it has G parameters, fixed as $m_i \rightarrow \infty$. For that model the quantities Eq. 34.5.1 evaluated at the individual observations are $q_i = 1/m_{g(i)}$, with $g(i)$ the group of observation i , so (G3) holds; and the information in group g is $m_g b''(\theta_g)/\phi \rightarrow \infty$, so (G4) holds. Part (a) applied to the fitted model against the saturated one, with $q = G - p$, gives $D/\phi \rightarrow \chi^2(G - p)$. For X^2 , a third-order expansion of d_i about $\hat{\mu}_i = y_i$ gives $w_i(y_i - \hat{\mu}_i)^2/V(\hat{\mu}_i) + O(w_i|y_i - \hat{\mu}_i|^3)$; with $w_i = m_i$ and $y_i - \hat{\mu}_i = O_p(m_i^{-1/2})$ the remainder summed over the G groups is $O_p(\max_i m_i^{-1/2})$, so $X^2 - D \rightarrow 0$ in probability.
- (c) For a binary response with the logit link, $\mathcal{M} = (0, 1)$ and $y_i \in \{0, 1\}$ is not in $\mathcal{M}$; taking limits in Eq. 34.5.4 with $0 \log 0 = 0$ gives the saturated log-likelihood 0 and $D = -2 \sum_i \{y_i \log \hat{\mu}_i + (1 - y_i) \log(1 - \hat{\mu}_i)\} = -2\ell(\hat{\beta})$. Writing $\hat{\eta}_i = \mathbf{x}_{(i)}^\top \hat{\beta}$ and using $\log(1 - \hat{\mu}_i) = -\log(1 + e^{\hat{\eta}_i})$,

$$\ell(\hat{\beta}) = \sum_i \{y_i \hat{\eta}_i - \log(1 + e^{\hat{\eta}_i})\} = \hat{\beta}^\top \mathbf{X}^\top \mathbf{y} - \sum_i \log(1 + e^{\mathbf{x}_{(i)}^\top \hat{\beta}}).$$

By Corollary 34.3.2, $\mathbf{X}^\top \mathbf{y} = \mathbf{X}^\top \hat{\mu}$, so $\ell(\hat{\beta}) = \hat{\beta}^\top \mathbf{X}^\top \hat{\mu} - \sum_i \log(1 + e^{\mathbf{x}_{(i)}^\top \hat{\beta}})$ depends on the data only through $\hat{\beta}$, and two data sets with the same $\hat{\beta}$ have the same deviance.

■

Example. (Grouped and ungrouped, simulated)

A logistic model with two parameters was fitted to 2000 data sets simulated from it, in two formats with the same number of Bernoulli trials over the same range of x : as 40 groups of 20 trials, and as 800 single trials. (The listing below repeats the experiment with 200 replicates, so that it runs in a few seconds.)

For the grouped data $G - p = 38$, and the simulated deviances have mean 39.74 and standard deviation 9.43, against the $\chi^2(38)$ values 38 and $\sqrt{76} = 8.72$: usable. For the ungrouped data $n - p = 798$, and the simulated deviances have mean 987.95 and standard deviation 20.62, against the $\chi^2(798)$ values 798 and 39.95 — the mean off by

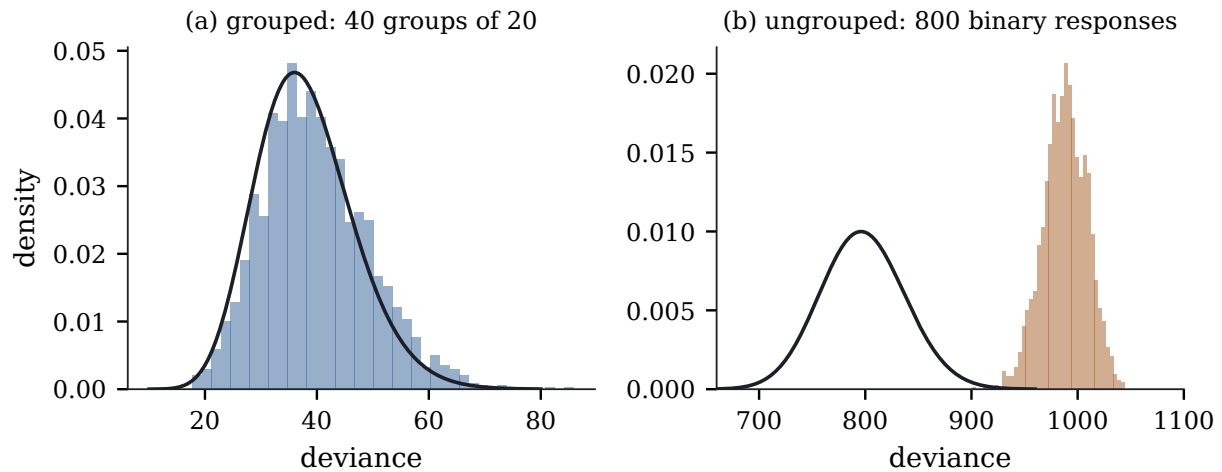


Figure 34.5.2: Simulated null distributions of the residual deviance, with the χ^2 density that the naive degrees-of-freedom count suggests. (a) Grouped data: the approximation works. (b) The same information as ungrouped binary responses: it does not.

nearly 200, the spread by a factor of two (Figure 34.5.2). Referring that deviance to $\chi^2(n - p)$ would reject a correct model with overwhelming confidence.

Nothing is wrong with the sample size. The two designs carry very nearly the same Fisher information — the script checks that the standard errors of $\hat{\beta}$ agree to within 2% — so the estimates behave alike. What fails is the reference distribution, because the saturated model in the second format has 800 parameters and grows with the data.

```
def logit_irls(X, y, w, steps=25):
    """Maximum likelihood for a binomial model with logit link; y is a proportion.
    """
    mu = (w * y + 0.5) / (w + 1.0)
    eta = np.log(mu / (1 - mu))
    beta = np.zeros(X.shape[1])
    for _ in range(steps):
        d = mu * (1 - mu)
        W = w * d
        z = eta + (y - mu) / d
        beta = np.linalg.solve((X.T * W) @ X, (X.T * W) @ z)
        eta = X @ beta
        mu = 1 / (1 + np.exp(-eta))
    return beta, mu

def binomial_deviance(y, mu, w):
    t1 = np.where(y > 0, y * np.log(np.where(y > 0, y, 1.0) / mu), 0.0)
    t2 = np.where(y < 1, (1 - y) * np.log(np.where(y < 1, 1 - y, 1.0) / (1 - mu)),
    0.0)
    return 2 * np.sum(w * (t1 + t2))

def null_deviances(X, beta, m, reps, rng):
    """Residual deviance of the fitted model, on `reps` data sets simulated from it.
    """
    p = 1 / (1 + np.exp(-(X @ beta)))
    w = np.full(len(p), float(m))
    out = np.empty(reps)
```

```

for r in range(reps):
    y = rng.binomial(m, p).astype(float) / m
    _, mu = logit_irls(X, y, w)
    out[r] = binomial_deviance(y, mu, w)
return out

rng = np.random.default_rng(3407)
beta0 = np.array([0.3, 0.9])
G, m = 40, 20                                # grouped: 40 groups of 20
        trials
Xg = np.column_stack([np.ones(G), np.linspace(-1.5, 1.5, G)])
N = 800                                       # ungrouped: 800 single trials
Xu = np.column_stack([np.ones(N), np.linspace(-1.5, 1.5, N)])

reps = 200
Dg = null_deviances(Xg, beta0, m, reps, rng)
Du = null_deviances(Xu, beta0, 1, reps, rng)
print(f"grouped    : mean {Dg.mean():8.2f} (df {G - 2}), sd {Dg.std():6.2f}"
      f" (sqrt(2 df) {np.sqrt(2 * (G - 2)):.2f})")
print(f"ungrouped  : mean {Du.mean():8.2f} (df {N - 2}), sd {Du.std():6.2f}"
      f" (sqrt(2 df) {np.sqrt(2 * (N - 2)):.2f})")

```

[Run this code online](#)

Idea. Differences of deviances between nested models are reliable, and are what the analysis-of-deviance table uses. The deviance *itself* is a goodness-of-fit statistic only for grouped data with large groups. The two claims are constantly confused because software prints them in the same table.

The analysis-of-deviance table is built like the sequential analysis-of-variance table of Definition 9.3.1: add terms in a chosen order, recording the drop in deviance at each step with its degrees of freedom. The reference distribution is chi-squared when ϕ is known and F when it is estimated.

Example. (Analysis of deviance for Engel's data)

Continue the gamma/log model of Example (Food expenditure: three models, two decisions), with u the centred logarithm of income and $n = 235$. Fitting the three nested models gives

model	residual df	deviance	change	F	p
1	234	39.01			
1 + u	233	4.27	34.73	1975.2	$< 10^{-4}$
1 + $u + u^2$	232	4.18	0.0918	5.22	0.023

Here ϕ is unknown, so each change is divided by $\hat{\phi} = 0.01758$ and referred to $F(1, 232)$, following [Theorem 34.5.5\(a\)](#). One qualification: the log link is not the gamma's canonical link, so (G1) does not hold and the calibration rests not on the proof given but on [Proposition 34.5.2](#), which extends the limit to this case without proving it. Income matters overwhelmingly; the quadratic term is borderline. The residual deviance 4.18 is *not* to be compared with $\chi^2(232)$: the data are ungrouped, so [Theorem 34.5.5\(b\)](#) does not apply, and the small value reflects the small dispersion of the gamma fit, not a suspiciously good one.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

engel = sm.datasets.engel.load_pandas().data
y = engel["foodexp"].to_numpy()
u = np.log(engel["income"].to_numpy())
u = u - u.mean()                                # centred, so the terms are less
                                                collinear
n = len(y)

def gamma_deviance(y, mu):
    """D = 2 sum { -log(y/mu) + (y - mu)/mu } for the gamma family."""
    return 2 * np.sum(-np.log(y / mu) + (y - mu) / mu)

models = {"1": np.ones((n, 1)),
          "1 + u": np.column_stack([np.ones(n), u]),
          "1 + u + u^2": np.column_stack([np.ones(n), u, u ** 2])}
fits, dev = {}, {}
for name, Xm in models.items():
    f = sm.GLM(y, Xm, family=sm.families.Gamma(sm.families.links.Log())).fit()
    fits[name], dev[name] = f, gamma_deviance(y, f.fittedvalues)

full = fits["1 + u + u^2"]
mu_full = full.fittedvalues
phi_pearson = np.sum((y - mu_full) ** 2 / mu_full ** 2) / (n - 3)
phi_deviance = dev["1 + u + u^2"] / (n - 3)

names = list(models)
print(f"{'term':14s} {'df':>4s} {'deviance':>10s} {'change':>9s} {'F':>8s} {'p'
      '':>8s}")
prev = None
for name in names:
    df_res = n - models[name].shape[1]
    row = f"{'name':14s} {'df_res':4d} {'dev[name]:10.4f}"
    if prev is not None:
        change = dev[prev] - dev[name]
        F = change / phi_pearson
        row += f" {'change':9.4f} {'F':8.2f} {'stats.f.sf(F, 1, n - 3):8.4f}"
    print(row)
    prev = name
print(f"dispersion: Pearson {phi_pearson:.5f}    deviance {phi_deviance:.5f}")
```

[Run this code online](#)

34.5.5 Estimating the dispersion

For the binomial and Poisson families $\phi = 1$ by construction; if the data contradict that, the remedy is a different model, which is [Chapter 38](#). For the normal, gamma and inverse Gaussian families ϕ is free. Maximum likelihood would take it from the full likelihood, which for the gamma involves the digamma function and behaves badly under misspecification. The standard choice is instead the **Pearson estimate**

$$\hat{\phi} = \frac{X^2}{n-p} = \frac{1}{n-p} \sum_{i=1}^n \frac{w_i(y_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)}. \quad (34.5.6)$$

The motivation is the moment identity $E\{w_i(Y_i - \mu_i)^2/V(\mu_i)\} = \phi$, exact at the true mean; replacing μ_i by $\hat{\mu}_i$ costs p degrees of freedom, by analogy with $s^2 = \text{SSE}/(n-p)$, the exact normal-theory case (Theorem 5.6.1). The alternative $D/(n-p)$ is also in use; for Engel's data the two give 0.01758 and 0.01803. They can differ sharply for skewed data, and the Pearson version is the more reliable, since it does not depend on the far tail of the assumed density.

Because the likelihood equations do not involve ϕ (Theorem 34.3.1(a)), estimating it does not disturb $\hat{\beta}$. It does enter every standard error, through $\hat{\mathbf{A}}_n^{-1} = \hat{\phi}(\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1}$, and its own variability is why the tests are then referred to an F distribution.

34.5.6 Residuals and diagnostics

The raw residual $y_i - \hat{\mu}_i$ is uninformative on its own, because its variance depends on $\hat{\mu}_i$. Three standardizations are in use.

Definition 34.5.2 — Pearson, deviance and Anscombe residuals

For a fitted generalized linear model, with d_i the deviance contributions of [Eq. 34.5.4](#) and h_{ii} the diagonal of the weighted hat matrix $\mathbf{H} = \hat{\mathbf{W}}^{1/2} \mathbf{X}(\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1} \mathbf{X}^\top \hat{\mathbf{W}}^{1/2}$, define

$$\begin{aligned} r_i^P &= \frac{\sqrt{w_i}(y_i - \hat{\mu}_i)}{\sqrt{\hat{\phi} V(\hat{\mu}_i)}}, & r_i^D &= \text{sign}(y_i - \hat{\mu}_i) \sqrt{d_i/\hat{\phi}}, \\ r_i^A &= \frac{\sqrt{w_i}\{A(y_i) - A(\hat{\mu}_i)\}}{\sqrt{\hat{\phi} A'(\hat{\mu}_i) \sqrt{V(\hat{\mu}_i)}}}, \end{aligned}$$

where $A(\mu) = \int^\mu V(t)^{-1/3} dt$. The **standardized** versions divide by $\sqrt{1 - h_{ii}}$.

The Pearson residuals have squares adding to $X^2/\hat{\phi}$, the deviance residuals squares adding to $D/\hat{\phi}$; each splits its own discrepancy measure into per-observation pieces, so a large deviance residual marks an observation the likelihood explains badly. The Anscombe residual first applies the transformation that brings the family closest to normal, $A(\mu) = \frac{3}{2}\mu^{2/3}$ for the Poisson and $3\mu^{1/3}$ for the gamma.

Example. (Which residual is closest to normal)

Two thousand observations were simulated from a Poisson log-linear model with fitted

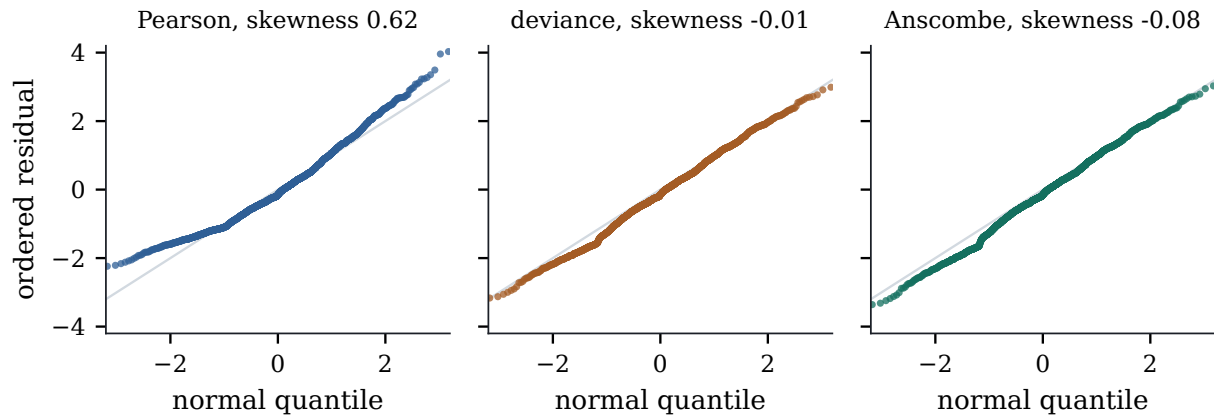


Figure 34.5.3: Normal quantile plots of the three residual types for a Poisson log-linear fit with small means. The reference line is the identity. Pearson residuals inherit the skewness of the family; deviance and Anscombe residuals largely remove it.

means between about 1 and 6, and the model refitted. Figure 34.5.3 shows normal quantile plots of the three residual types. The Pearson residuals are visibly right-skewed, with sample skewness 0.618 — the skewness $\mu^{-1/2}$ of Exercise (B3), averaged over the fitted means. The deviance residuals have skewness -0.010 and the Anscombe residuals -0.081 ; both are far closer to normal, and neither is uniformly better. Discreteness remains visible in all three as short horizontal runs. The leverages sum to 2 (Proposition 6.8.1(b)) and the largest is 0.0029.

```

rng = np.random.default_rng(3405)
N = 2000
Xp = np.column_stack([np.ones(N), rng.uniform(-1, 1, N)])
mu_true = np.exp(Xp @ np.array([1.0, 0.8])) # means between about 1
and 6
yp = rng.poisson(mu_true).astype(float)
pois = sm.GLM(yp, Xp, family=sm.families.Poisson()).fit()
mu_p = pois.fittedvalues

pearson = (yp - mu_p) / np.sqrt(mu_p) # r = (y -
mu)/sqrt(V)
d_i = 2 * (np.where(yp > 0, yp * np.log(np.maximum(yp, 1e-300) / mu_p), 0.0) - (yp -
mu_p))
deviance_r = np.sign(yp - mu_p) * np.sqrt(d_i) # signed
root of d_i
anscombe = 1.5 * (yp ** (2 / 3) - mu_p ** (2 / 3)) / mu_p ** (1 / 6) # A(mu) =
1.5 mu^{2/3}

W = mu_p # working weights for
Poisson/log
root = np.sqrt(W)[: , None] * Xp
H = root @ np.linalg.inv(root.T @ root) @ root.T # the weighted hat matrix
h = np.diag(H)

for name, r in [("Pearson", pearson), ("deviance", deviance_r), ("Anscombe",
anscombe)]:
    print(f"{name:9s} sd {r.std(ddof=2):6.3f} skewness {stats.skew(r):7.3f}")
print(f"leverages: sum {h.sum():.4f}, largest {h.max():.4f}")

```

Run this code online

The rest of [Chapter 20](#) transfers under the substitution $\mathbf{X} \mapsto \hat{\mathbf{W}}^{1/2}\mathbf{X}$, which is Pregibon's (1981) extension of the linear-model diagnostics: the leverages of Proposition 6.8.1, a Cook-type measure $C_i = (r_i^P)^2 h_{ii} / \{p(1-h_{ii})^2\}$ (Definition 20.4.1), and the one-step deletion approximations of Theorem 20.3.1 with the working response in place of $\mathbf{y}$. The exact deletion identities do not, because dropping a case changes the weights too. For ungrouped binary data the residuals take only two values at each $\hat{\mu}_i$, so only smoothed plots carry a signal — a point Proposition 35.4.1 develops.

34.5.7 Model selection

Because the fit is by maximum likelihood, Definition 29.2.2 applies directly, with $\text{AIC} = -2\ell(\hat{\boldsymbol{\beta}}) + 2d$ and $d = p$ when ϕ is known, $p + 1$ when it is estimated; for a $\phi = 1$ family comparing AIC is comparing $D + 2p$. The rest of [Chapter 29](#) transfers unchanged, Proposition 29.5.1 included.

34.5.8 Exercises

34.5.8.1 A. Check your understanding

Exercise A1.

Verify the normal and Poisson rows of the deviance table from [Eq. 34.5.4](#). For the Poisson, show that if the model has an intercept then $\sum_i (y_i - \hat{\mu}_i) = 0$ by Corollary 34.3.2, so the deviance reduces to $2 \sum_i y_i \log(y_i / \hat{\mu}_i)$.

Exercise A2.

A binomial model is fitted to 60 groups with 4 parameters, and the residual deviance is 71.2 on 56 degrees of freedom. When is comparing it with $\chi^2(56)$ legitimate, and what would you need to know about the data first?

Solution. By [Theorem 34.5.5\(b\)](#) the comparison needs the number of groups fixed and the group sizes large, so one needs the m_i : if they are all in the hundreds the comparison is reasonable, and if many are 1 or 2 it is not, whatever the total sample size. A rough rule is that the expected counts $m_i \hat{\mu}_i$ and $m_i(1 - \hat{\mu}_i)$ should all exceed about five.

34.5.8.2 B. Practice

Exercise B1.

Let $\hat{\beta}$ have asymptotic standard error s . Compute the Wald statistic for $H_0 : \beta = 0$ and, using the delta method, the Wald statistic for the equivalent hypothesis $H_0 : e^\beta = 1$. Show that their ratio is $\{(e^{\hat{\beta}} - 1)/\hat{\beta}\}^2 e^{-2\hat{\beta}}$, and evaluate it at $\hat{\beta} = 1$ and $\hat{\beta} = 3$. Why is the likelihood ratio statistic unaffected?

Exercise B2.

Show that $E\{w_i(Y_i - \mu_i)^2/V(\mu_i)\} = \phi$ exactly, for every observation in an exponential dispersion family. Deduce that X^2/n is unbiased for ϕ when the true means are used, and explain why $X^2/(n-p)$ is only approximately unbiased when they are estimated.

Exercise B3.

Derive the Anscombe residual for the Poisson family from $A(\mu) = \int^\mu t^{-1/3} dt$, and check that the standardizing denominator is $\hat{\mu}^{1/6}$. Explain why A is *not* the variance-stabilizing transformation $2\sqrt{\mu}$ of Corollary 22.4.2, and what each one is optimizing.

Solution. $A(\mu) = \frac{3}{2}\mu^{2/3}$, so $A'(\mu) = \mu^{-1/3}$ and $A'(\hat{\mu})\sqrt{V(\hat{\mu})} = \hat{\mu}^{-1/3}\hat{\mu}^{1/2} = \hat{\mu}^{1/6}$, giving $r^A = \frac{3}{2}(y^{2/3} - \hat{\mu}^{2/3})/\hat{\mu}^{1/6}$. The variance-stabilizing transformation solves $A' \propto V^{-1/2}$ and makes the *variance* constant; the Anscombe transformation solves $A' \propto V^{-1/3}$ and makes the *third cumulant*, hence the skewness, vanish to the order considered. Different objectives, different exponents; the Anscombe residual is then standardized by hand, so it does not need the variance to be stable by itself.

Exercise B4.

Two summaries of predictive power for a generalized linear model are the sample correlation between $\mathbf{y}$ and $\hat{\boldsymbol{\mu}}$, and the likelihood-based $R_L^2 = (\ell_M - \ell_0)/(\ell_{\text{sat}} - \ell_0)$, where ℓ_0 , ℓ_M and ℓ_{sat} are the maximized log-likelihoods of the intercept-only model, the fitted model and the saturated model. Show that for a family with $\phi = 1$ the second is $(D_0 - D)/D_0$, with D_0 the null deviance, and deduce that it lies in $[0, 1]$ and never decreases when a term is added. Why does that last property make it a poor basis for choosing a model?

Solution. With $\phi = 1$, Definition 34.5.1 gives $D = 2(\ell_{\text{sat}} - \ell_M)$ and $D_0 = 2(\ell_{\text{sat}} - \ell_0)$, so $\ell_M - \ell_0 = (D_0 - D)/2$ and $R_L^2 = (D_0 - D)/D_0$. Both deviances are nonnegative by Proposition 34.5.4 and $D \leq D_0$, since enlarging a model cannot lower the maximized likelihood; so $R_L^2 \in [0, 1]$ and cannot decrease when a term is added. That is the defect of the unadjusted R^2 of Section 9.5, the point Proposition 9.5.3 makes there: it rewards complexity unconditionally. AIC instead charges 2 per parameter against -2ℓ , and so can rise.

34.5.8.3 C. Going deeper**Exercise C1.**

Extend Theorem 34.5.5(c) to a Poisson log-linear model in which every y_i is 0 or 1: show that the deviance is again a function of $\boldsymbol{\beta}$ alone. Then show that the argument breaks down as soon as some $y_i \geq 2$, and explain in words what extra information the larger counts supply. (This is the reason the deviance is a serviceable goodness-of-fit statistic for Poisson data with large means and a useless one for sparse counts.)

Solution. For the Poisson deviance, $D = 2 \sum_i \{y_i \log(y_i/\hat{\mu}_i) - (y_i - \hat{\mu}_i)\}$. When every $y_i \in \{0, 1\}$ the term $y_i \log y_i$ is zero, so $D = -2 \sum_i y_i \log \hat{\mu}_i + 2 \sum_i (y_i - \hat{\mu}_i)$; the second sum vanishes by Corollary 34.3.2 if the model has an intercept, and the first is $-2\hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y} = -2\hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \hat{\boldsymbol{\mu}}$, again by Corollary 34.3.2. So D is a function of $\hat{\boldsymbol{\beta}}$. If some $y_i \geq 2$, the term $\sum_i y_i \log y_i$ is a nonzero function of the data that is *not* determined by $\mathbf{X}^\top \mathbf{y}$: two data sets with the same sufficient statistic, one with counts $(2, 0)$ at two identical covariate values and one with $(1, 1)$, have the same $\hat{\boldsymbol{\beta}}$ but different deviances. The larger counts carry information about the spread of the responses within a covariate pattern, which is precisely what a goodness-of-fit statistic needs and what binary data cannot supply.